

PYTHON ASSIGNMENT

SKILL TEST-1

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MCQ'S

Question 1

Which of the following best describes a loop in Python?

- A) A condition checker
- B) A repetition control structure
- C) A function block
- D) A variable declaration

Answer: B

Explanation: Loops execute a block of code repeatedly based on a condition.

Question 2

Write a Python code snippet that checks if a number is even or odd using if-else statement.

- A) `if num % 2 == 0: print('Even') else: print('Odd')`
- B) `if num % 2 != 0: print('Even') else: print('Odd')`
- C) `if num % 2: print('Even') else: print('Odd')`
- D) `if num % 2 == 1: print('Even') else: print('Odd')`

Answer: A

Question 3

What will be the output of the following code?

```
for i in range(2):  
    for j in range(2):  
        for k in range(2):  
            print(i, j, k, end=" ")
```

- A) 000010100011100101111
- B) 000001010011100101111
- C) 000001010011100101110111
- D) 000001010011100111111

Correct Output:

000001010011100101111

Answer: A

Question 4

What will be the output of the following code?

```
for i in range(1, 4):  
    for j in range(i):  
        print("*", end=" ")  
    print()
```

- | | | | |
|------|--------|------|-------|
| A) * | B) *** | C) * | D) ** |
| ** | ** | *** | * |
| *** | * | ** | *** |

Answer: A

Explanation: Nested loops print stars in decreasing and then increasing order.

Question 5

What will be the output of the following code?

```
s = "Python"  
print(s.find("y"))
```

- A) 0
- B) 1
- C) -1
- D) Error

Answer: B

Explanation: "Python".find("y") returns index 1.

Problem Statement-1

Ravi is calculating a modular arithmetic-based series to test a number system in his new program. For a given integer n , he computes the sum of terms in the form $(i^2 + 1) \% m$ for values of i from 1 to $\min(n, 5)$. Write a program to compute and output the sum of the first up to 5 such terms based on n and modulus m .

Example

Input: **Output:**

5 18

7

Explanation

Compute each term of the series:

- $a1 = (1^2 + 1) \% 7 = 2$
- $a2 = (2^2 + 1) \% 7 = 5$
- $a3 = (3^2 + 1) \% 7 = 3$
- $a4 = (4^2 + 1) \% 7 = 3$
- $a5 = (5^2 + 1) \% 7 = 5$

Compute the sum of the terms:

- $\text{Sum} = 2 + 5 + 3 + 3 + 5 = 18$

Input format :

The first line of input consists of integer n representing the number of terms in the series.

The next line of input consists of integer m representing the modulus.

Output format :

The output prints an integer representing the computed sum of the series

Sample test cases :

Input 1 :

5

Output 1 :

18

7

Input 2 :

3

6

Output 2 :

11

CODE:

```
n = int(input())
m = int(input())

a1 = (1**2 + 1) % m
a2 = (2**2 + 1) % m if n >= 2 else 0
a3 = (3**2 + 1) % m if n >= 3 else 0
a4 = (4**2 + 1) % m if n >= 4 else 0
a5 = (5**2 + 1) % m if n >= 5 else 0
sum_of_series = a1 + a2 + a3 + a4 + a5
print(sum_of_series)
```

Problem Statement-2

David, a sports manager, is standardizing team names in his database. He wants each word formatted with the first letter in uppercase and the rest in lowercase, regardless of the input case. The program should ensure only a single space remains between words, removing any leading, trailing, or multiple spaces. Additionally, David needs to count the total number of words and find the length of the longest word.

Help David by creating a program that formats the team name, counts the words, and identifies the longest word's length.

Input format :

The input consists of a string representing the team name.

Output format :

The first line of output prints the formatted team name as a string.

The second line of output prints the total number of words as an integer.

The third line of output prints the length of the longest word as an integer.

Sample test cases :

Input 1 :

los angeles lakers

Output 1:

Los Angeles Lakers

3

7

Input 2:

new york GIANTS

Output 2 :

New York Giants

3

6

CODE:

```
team_name = input()
words = team_name.strip().split()
formatted_name = ""
max_word_length = 0
for word in words:
    formatted_word = word[0].upper() + word[1:].lower()
    if formatted_name:
        formatted_name += " "
    formatted_name += formatted_word
    word_length = len(formatted_word)
    if word_length > max_word_length:
        max_word_length = word_length
word_count = len(words)
print(formatted_name)
print(word_count)
print(max_word_length)
```

Problem Statement-3

In a fitness challenge program, participants are rated based on the number of hours they exercise in a week. The ratings are based on the following criteria:

- Overachiever: More than 25 hours
- Excellent: 20 to 25 hours
- Good: 15 to 19 hours
- Average: 10 to 14 hours
- Needs Improvement: 5 to 9 hours
- Poor: Below 5 hours

Additionally, there are some performance bonus multipliers:

- Overachiever: 1.2x multiplier
- Excellent: 1.1x multiplier
- Good: 1.05x multiplier

The final score for each participant will be printed along with their rating.

Input format :

The first line contains an integer N, representing the number of participants

The next N lines each contain a decimal value representing the hours of exercise for each participant in a week.

Output format :

For each participant, the output prints a line containing the rating and final score separated by a space. The final score is rounded to two decimal places.

If the hours are negative or exceed 168, the output prints "Invalid input: " followed by the hours value rounded to two decimal places.

Input 1:

3

22.5

18.3

3.2

Output 1:

Excellent 24.75

Good 19.21

Poor 3.20

Input 2:

2

27.5

-2.0

Output 2:

Overachiever 33.00

Invalid input: -2.00

CODE:

```
n=int(input())
for i in range(n):
    a=float(input())
    if a>=0 and a<=168:
        if a>25:
            s=1.2*a
            print(f"Overachiever {s:.2f}")
        elif a>=20 and a<=25:
            s=1.1*a
            print(f"Excellent {s:.2f}")
        elif a>=15 and a<=19:
            s=1.05*a
            print(f"Good {s:.2f}")
        elif a>=10 and a<=14:
            print(f"Average {a:.2f}")
        elif a>=5 and a<=9:
            print(f"Needs Improvement {a:.2f}")
        else:
            print(f"Poor {a:.2f}")
    else:
        print(f"Invalid input: {a:.2f}")
```